Java String Operations

15 separate Java programs with brief explanations and console-output screenshots.

2. Find String Length

The length() method returns the number of characters in a string, including spaces.

```
class StringLength {
    public static void main(String[] args) {
        String text = "Hello";
        System.out.println(text.length());
    }
}
```

```
Console output
> 5
```

3. Concatenate Strings

The + operator joins two strings to form one longer string.

```
class ConcatenateString {
    public static void main(String[] args) {
        String first = "Hello";
        String second = " World";
        System.out.println(first + second);
    }
}
```

```
Console output
> Hello World
```

4. Compare Strings

equals() compares the text stored in two strings and returns true when they match.

```
class CompareString {
    public static void main(String[] args) {
        String a = "Java";
        String b = "Java";
        System.out.println(a.equals(b));
    }
}
```

```
Console output
> true
```

5. Convert to Uppercase

toUpperCase() changes all lowercase letters in a string to uppercase letters.

```
class UppercaseString {
    public static void main(String[] args) {
        String text = "hello";
        System.out.println(text.toUpperCase());
    }
}
```

```
Console output
> HELLO
```

6. Convert to Lowercase

toLowerCase() changes all uppercase letters in a string to lowercase letters.

```
class LowercaseString {
    public static void main(String[] args) {
        String text = "HELLO";
        System.out.println(text.toLowerCase());
    }
}
```

```
Console output
> hello
```

7. Extract a Substring

substring(start, end) extracts characters from start up to, but not including, end.

```
class SubstringExample {
    public static void main(String[] args) {
        String text = "Programming";
        System.out.println(text.substring(0, 7));
    }
}
```

```
Console output
> Program
```

8. Find Character Position

indexOf() returns the first position of a character. Positions start from 0.

```
class IndexOfExample {
    public static void main(String[] args) {
        String text = "Hello";
        System.out.println(text.indexOf('o'));
    }
}
```

```
Console output
> 4
```

9. Replace Text

replace() changes every occurrence of the specified text with new text.

```
class ReplaceString {
    public static void main(String[] args) {
        String text = "Java is fun";
        System.out.println(text.replace("fun", "powerful"));
    }
}
```

```
Console output
> Java is powerful
```

10. Split a String

split() divides a string into an array using a separator. This program prints each array item.

```
class SplitString {
    public static void main(String[] args) {
        String text = "Java,Python,C++";
        String[] languages = text.split(",");

        for (String language : languages) {
            System.out.println(language);
        }
    }
}
```

```
●●● Console output
> Java
> Python
> C++
```

11. Join Strings

String.join() combines multiple strings by placing the chosen separator between them.

```
class JoinString {
    public static void main(String[] args) {
        String result = String.join("-", "Java", "String", "Methods");
        System.out.println(result);
    }
}
```

```
●●● Console output
> Java-String-Methods
```

12. Trim Spaces

trim() removes whitespace from the beginning and end of a string.

```
class TrimString {
    public static void main(String[] args) {
        String text = "  Hello Java  ";
        System.out.println(text.trim());
    }
}
```

```
●●● Console output
> Hello Java
```

13. Reverse a String

StringBuilder.reverse() reverses the characters. toString() converts the result back to a String.

```
class ReverseString {
    public static void main(String[] args) {
        String text = "Hello";
        String reversed = new StringBuilder(text).reverse().toString();
        System.out.println(reversed);
    }
}
```

```
●●● Console output
> olleH
```

14. Check Starts With

startsWith() checks whether a string begins with specified text and returns true or false.

```
class StartsWithString {
    public static void main(String[] args) {
        String text = "Hello Java";
        System.out.println(text.startsWith("Hello"));
    }
}
```

```
Console output
> true
```

15. Check Ends With

endsWith() checks whether a string ends with specified text and returns true or false.

```
class EndsWithString {
    public static void main(String[] args) {
        String text = "Hello Java";
        System.out.println(text.endsWith("Java"));
    }
}
```

```
Console output
> true
```

16. Check Whether a String Is Empty

isEmpty() returns true when a string contains zero characters.

```
class EmptyString {
    public static void main(String[] args) {
        String text = "";
        System.out.println(text.isEmpty());
    }
}
```

```
Console output
> true
```